

THEME: MEDIA, CONTENT & DIGITAL EXPERIENCES

ContextBridge AI

"Not a tutor. A compass for self-learners — every answer anchored to the video you chose, in your language, with proof, and honest about where the video ends."

AI REASONING ENGINE

Gemini 2.5 Flash

Multimodal video analysis,
native audio transcription,
and Google Search grounding.

UNSUPPORTED ANSWERS

0.0%

Strict Anti-Fabrication Gate:
Verbatim transcript quote
matching & exact timestamps.

PRODUCTION DEPLOYMENT

Google Cloud Run

Next.js 16 Web + FastAPI
Stateless containerized stack
Serving 100% live traffic.

The Self-Learner's Dilemma

Billions of hours of educational video exist, but self-directed learners face four critical friction points.

1. Scrubbing Fatigue (Wasted Time)

Learners spend 15+ minutes hunting back and forth across 45-minute tutorials just to locate a 10-second syntax snippet or formula.

2. Chatbot Hallucination Trap

Generic chatbots (ChatGPT, Claude) lack the video's actual runtime. They invent plausible-sounding code and quote words never spoken.

3. The Linguistic Barrier

World-class technical tutorials are overwhelmingly in English.
Non-native speakers (Hindi, Hinglish) struggle with dense video jargon.

4. The Blurry Knowledge Boundary

Existing summarizers fail to distinguish between what the creator said and general web knowledge, completely eroding learner confidence.

The Three Pillars of The Compass

ContextBridge enforces trust through structural, mathematical gates.

■ P1

Anchored

The Video is Sole Ground Truth

- Video answers must originate from the creator's transcript.
- `quote_matches_transcript` gate strictly verifies character match.
- Invariant: zero invented quotes.

■ P2

Proof

Trust Visible in 10 Seconds

- EvidenceBadge displays origin and verification confidence.
- Click seeks player to exact second (~1s precision).
- Auto-scrolls transcript segment.

■ P3

Boundary Honesty

Honesty is a Feature

- Video limits are acknowledged.
- External research framed in a dedicated BoundaryCard.
- Real Google Search citations.
- Never blends creator & web claims.

Trust Visible: Interactive Workspace

A cohesive UI delivering synchronized proof at every touchpoint.

EvidenceBadge

3 Explicit States:

- Verified Video · 98%
- Beyond Video (Web)
- Honest Boundary

TranscriptSync

Bidirectional synchronization:

Clicking an evidence quote
auto-scrolls right-rail
transcript to active segment.

ExploreSuggestions

"Explore from here →" chips
derived dynamically from
video chapters & topics.
Zero extra LLM latency!

Contradictions

Critical thinking helper:
Highlights internal nuance
with dual clickable jump
buttons (00:05 vs 00:15).

■ BoundaryCard: Explicit Separation of Knowledge Worlds

When questions exceed video scope, external research is isolated in a distinct sky-blue frame with real Google Search sources.

Bilingual Voice Loop (Hindi, Hinglish, English)

Learners explore in the language they think in — full voice-in, voice-out parity with Gemini.

■ Speech-to-Text

Gemini 2.5 Flash native audio
transcription via POST /transcribe.
Verbatim capture of Hindi/Hinglish.

■ Auto-Speak & TTS

Language-matched speech
synthesis (hi-IN / en-US).
Hands-free audio self-learning.

■ Instant Barge-In

Tapping mic cancels active audio.
Animated avatar rings and audio
waveforms indicate speaking state.

LIVE VERIFIED BENCHMARK CASE (q13 & q14 from eval_results.json):

Student (Hindi): "■■■■■ ■■■■■■ ■■■■■■ ■■■ ■■■■ ■■■ ■■■■ ■■■■?"

ContextBridge: "Saeka Shimada Tokyo mein ek photographer hain (00:01)."

Verified Evidence: [00:01 - 00:03] "My name is Saeka Shimada. I'm a photographer in Tokyo."

Confidence: 0.98 | Timestamp Retrieval: EXACT (1.0s - 3.0s) | Latency: 2.35s

The Anti-Fabrication Agent Gate

A 3-tool bounded state machine with mathematical limits on LLM rounds.

// THE 3-TOOL BOUNDED LADDER

1. OBSERVE: `retrieve_video_context`

- Lexical token matching over chapters & transcript
- Devanagari fallback window for Hindi queries

2. EVALUATE: `validate_answer_draft (GATE)`

- Validates $0 \leq \text{startSeconds} \leq \text{endSeconds} \leq \text{duration}$
- `quote_matches_transcript` verifies exact quote
- Exactly 1 retry on gate failure -> `declare_not_found`

3. SELECT: `gemini_web_research`

- Activates only when research enabled & not in video
- Returns verified Google Search grounding chunks

Strict Budget: ≤ 2 LLM Rounds

- Exactly 1 answering branch per question.
- Max 1 validation retry = guaranteed ≤ 2 LLM rounds.
- No infinite loops, no multi-agent token waste.
- Average response latency: 3.59s on live Cloud Run.

Stateless Cloud Run Design

- Client owns conversation history (last ≤ 6 turns).
- Server maintains zero session memory or database lock.
- Infinite horizontal scale across Google Cloud Run.
- Docker containerized Next.js 16 + FastAPI.

Live 25-Case Evaluation Benchmark

Continuously verified against the live Google Cloud Run deployment.

0.0%

Unsupported Answers

Target: 0.0% Strict

94.4%

Timestamp Accuracy

Target: $\geq 90.0\%$

92.0%

Groundedness Rate

Target: $\geq 90.0\%$

3.59s

Average Latency

Target: $< 5.0s$

Test Suite & Failure-Path Drills (Zero Failures Across All Suites):

- 83/83 Unit & Regression Tests Passing (pytest tests/) in 0.82 seconds
- 9/9 Architecture §15 Failure-Path Drills Passing (scripts/test_failure_paths.py)
- 100% Clean ESLint Gate (0 errors, 0 warnings) + Turbopack Production Build Clean
- Full P0-1 through P0-9 Acceptance Sweep Passing on Live Cloud Run (scripts/sweep_deployed.py)
- Eval Results persisted to context/eval_results.json (25 benchmark cases)

Why ContextBridge Wins

Not a passive summarizer, and not an ungrounded general-purpose chatbot.

Capability	Generic LLMs (ChatGPT / Claude)	Video Summarizers (NoteGPT)	ContextBridge AI
Verbatim Quote Validator	■ None (hallucinates quotes)	■ None (paraphrases only)	■ Enforced by validator
Sub-second Video Seeking	■ No video player link	■ Rough minute links	■ Exact second (~1s) seek
TranscriptSync Auto-Scroll	■ Absent	■ Static transcript text	■ Real-time auto-scroll
Contradiction Surfacing	■ Absent	■ Absent	■ Dual clickable timestamps
Strict BoundaryCard	■ Blends web & training	■ Fails on missing facts	■ Explicit Boundary frame
Bilingual Voice Loop	■ English bias / Text first	■ English only	■ Hindi/Hinglish TTS parity

Alignment with AI Builder Cup Criteria

Engineered to maximize evaluation score across all four criteria.

Technical Merit (40% Weight)

- Zero Hallucination: Structural anti-fabrication validator.
- Containerized Stack: Deployed to Google Cloud Run.
- 83 pytest tests + 9 automated failure-path drills.
- Multi-turn stateless conversation history passthrough.

Impact (25% Weight)

- Saves Hours: Cuts 45-minute scrubbing down to 1 second.
- Democratizes Learning: Hindi/Hinglish voice parity unlocks technical education for 600M+ non-native speakers.
- Accessibility: Adaptive explanation levels & WebVTT captions.

Innovation (25% Weight)

- The Compass Paradigm: Dynamic guided exploration chips.
- Epistemic Humility: Strict BoundaryCard for external research.
- Contradiction Surfacing: Critical thinking on conflicting claims.

UX & Presentation (10% Weight)

- 10-Second Trust: Instantly understandable EvidenceBadges.
- Fluid Animations: Synchronized transcript scroll & waveforms.
- Fast Seeking: 7-speed Radix dropdown + instant barge-in mic.

ContextBridge AI ■

The Trusted Compass for Self-Learners Worldwide.

■ LIVE WEB APP

contextbridge-web-
c5ltxo3mkq-uc.a.run.app

Try Tokyo lesson or upload clip

■ OPEN SOURCE GITHUB

github.com/
lvhNaruto/contextbridge-ai

Clean git history & full tests

■ INTERACTIVE API

contextbridge-api-
c5ltxo3mkq-uc.a.run.app/docs

OpenAPI 3.1 Swagger docs

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